Jot — Application Scope Document

March 26, 2026 • Pre-Release • macOS 26+ / iOS 26+ • SwiftUI + SwiftData

1. Executive Summary

Jot is a native macOS note-taking application built with SwiftUI, targeting macOS 26 and iOS 26. It features a rich text editor with inline media, folder-based organization, split-pane editing, voice recording with transcription, Apple Intelligence integration, and Apple's Liquid Glass design language.

The project has been in active development for 8 months (August 2025 to present) with 133 commits by a solo developer. The codebase is 45,703 lines of Swift across 109 files. All core features are implemented including card section blocks with rich text editing, tabs with rich text editing, AI-powered tools (translate, edit content, text generation), and a comprehensive code audit. A 13-feature roadmap is planned across 6 development batches.

2. Project Metrics

Metric	Value
Lines of Swift	45,703
Swift files	109
Git commits	133
Development span	8 months (Aug 2025 – Mar 2026)
Contributors	1 (Moheb Anwari)
SVG icon assets	100+
Semantic color tokens	28 (light + dark)
View components	37
Utility managers	19

3. Architecture

3.1 Project Structure

```
Jot/
  App/              JotApp.swift, ContentView.swift, NoteSelectionCommands.swift
  Models/           Note, Folder, Sticker, SearchEngine, CalloutData, CodeBlockData,
                    NoteTableData, TabsContainerData, CardSectionData
```

```
  SwiftData/      NoteEntity, FolderEntity, SimpleSwiftDataManager
Views/
  Screens/        NoteDetailView, NoteDetailView+Actions
  Components/     37 reusable UI components
Utils/            ThemeManager, FontManager, GlassEffects, TextFormattingManager,
                  SyntaxHighlighter, RichTextSerializer,
                  ImageStorageManager, NoteExportService, NoteImportService,
                  WebMetadataFetcher, BackupManager, NoteVersionManager,
                  AppleIntelligenceService, AudioRecorder, Transcriber, HapticManager,
                  KeychainManager, NoteAuthenticationManager, Extensions,
                  FloatingToolbarPositioner, TrafficLightAligner
Ressources/       Assets.xcassets (color sets, icon imagesets)
Jot.entitlements
Info.plist
```

3.2 Architectural Patterns

Pattern	Implementation
State management	@StateObject / @EnvironmentObject for NotesManager, ThemeManager, AuthManager
Persistence	SwiftData with SimpleSwiftDataManager wrapper; auto-save enabled
Data modeling	Dual-layer: lightweight Codable structs + @Model SwiftData entities
Rich text	NSTextView via NSViewRepresentable bridge, custom markup serialization
Block overlays	NSTextAttachment subclasses with overlay NSViews for tables, cards, tabs, callouts, code blocks
Design system	Liquid Glass (macOS 26+) with .ultraThinMaterial fallback
Performance	Batch loading (50/batch, 500 max), debounced auto-save (3s), cached sidebar groupings

3.3 Entitlements

Capability	Purpose
App Sandbox	Security isolation
User-Selected File R/W	Import/export files
Network Client	Web clip metadata fetching
Audio Input	Voice note recording
Speech Recognition	Audio transcription

4. Implemented Features

4.1 Note Management

Feature	Description
Create / Edit / Delete	Full CRUD with title, content, timestamps
Soft Delete & Trash	Notes move to trash with deletedDate; restore or permanent delete
Pinning	Pin notes to sidebar top; dedicated pinned section with flow layout
Archiving	Move notes to archive; separate archive view
Locking	Password-protect notes (custom password, system login, or Touch ID)
Tags	String array metadata on notes
Stickers	Emoji/text annotations with position, rotation, scale, z-index; 6 colors
Auto-save	Debounced (3s) persistence with draft snapshots

4.2 Rich Text Editor

Feature	Details
Text formatting	Bold, italic, underline, strikethrough, highlight
Headings	H1 (32pt), H2 (24pt), H3 (20pt)
Lists	Bullet, numbered, dashed, todo/checkbox
Text alignment	Left, center, right, justify
Text color	Custom hex color picker with recent colors
Highlight color	Custom hex highlight picker with recent colors
Indentation	Left/right indent controls
Block quotes	Styled quotation blocks
Font size	Configurable per-selection via floating toolbar
Font family	Charter, System (SF Pro), Monospaced per-selection
Squiggly strikethrough	Custom wavy line with content-seeded deterministic wobble, dimmed text at 0.7 alpha **NEW**
Undo/Redo	Native NSUndoManager integration
Find & Replace	In-note text search with match highlighting

4.3 Content Blocks

Block Type	Details
Card sections **NEW**	Multi-card blocks with per-card Tailwind colors (17 hues), individually resizable dimensions, vertical stacking into columns, drag reorder between/within columns, magnetic width and height snapping with visual guide bars, rich text editing with full formatting support, squiggly

	strikethrough, todo insertion, AI tool integration (translate, edit, generate), click-to-focus anywhere on card, custom layout manager, accessible via slash menu and toolbar
Tabs container UPDATED	Named tabs with 7 color options, resizable, add/remove/rename, now with full rich text editing support (bold, italic, headings, lists, block quotes, alignment, text/highlight color, font size/family, squiggly strikethrough, todo insertion, AI tool integration)
Code blocks	20+ languages, syntax highlighting, language selector, copy-to-clipboard
Callouts	5 types (Info, Warning, Tip, Note, Important) with icons and accent colors
Tables	Row/column CRUD, cell editing, resizable columns, text wrapping
Dividers	Horizontal rule insertion
Web clips	URL metadata (title, description, favicon), click to open
Inline images	Image picker with local storage, thumbnail caching
File attachments	Any file type, stored as references, reveal in Finder
Note links	Cross-note references via @ mention picker

4.4 Card Section Architecture NEW

Component	Details
Data model	`CardSectionData` with `columns: [[CardData]]` for 2D layout; flat-index bridge for backward compatibility
Serialization	`[[cards|col1spec;col2spec]]col1content;;col2content[[/cards]]` with legacy comma-format auto-conversion
Rich text	`RichTextSerializer` for tag-based inline serialization/deserialization shared with tabs; `TextFormattingManager` delegation
Layout manager	`_CardLayoutManager` — custom NSLayoutManager with squiggly strikethrough drawing, native strikethrough suppression
Notification routing	11 observers for formatting, todo, AI; `cardOrigin` flag prevents main editor double-handling; first-responder guards on main editor
Re-entrancy guard	`isSyncingContent` flag wraps struct mutations to prevent `didSet → syncTextViews()` feedback loops
Drag & drop	2D drop targeting: `.betweenColumns(Int)` and `.inColumn(column:Int, atRow:Int)`; bottom 35% = stack zone
Magnetic snap	Width snaps to sibling widths, height snaps to adjacent column bottom edges; 6pt threshold with CALayer guide bars

4.5 AI Tools NEW

Feature	Details

Translate	Selection-based translation with language picker; replaces in-place; works in main editor, cards, and tabs
Edit content	Custom instruction-based rewrites with before/after preview panel
Text generation	Prompt-based text generation inserted at cursor position
Overlay routing	`cardOrigin` flag threads through capture → NoteDetailView state → replacement notifications; overlay text views store `aiTargetCardTV` / `aiTargetTabTV` for post-focus-shift targeting

4.6 Command Menu UPDATED

Triggered by typing `/` in the editor. 23+ commands with arrow-key navigation and search filtering. Commands include: todo, image, code, callout, table, tabs, **cards**, headings (h1/h2/h3), bullet/numbered/dashed lists, blockquote, link, file, voice, divider, sticker.

Note picker triggered by `@` for cross-note linking with inline preview.

4.7 Folder & Organization

Feature	Details
Folder management	Create, rename, delete, archive folders
Folder colors	Custom hex color per folder
Drag to folder	Drag notes into folders in sidebar
Unfiled notes	Notes without folder assignment surface separately
Date grouping	Today, This Month, This Year, Older sections
Collapsible sections	Pinned, Locked, Folders, Archive

4.8 Search

Full-text search with relevance scoring: title (+100), folder name (+120), tags (#hashtag), content (+10). Debounced at 250ms. Stores up to 3 recent queries. Max 20 results with context snippets. Floating search UI with Liquid Glass styling.

4.9 Split View

Side-by-side editing of two notes. Adjustable split ratio via draggable divider. Drag a note to left/right zone to open split. Panes are swappable. Custom wavy divider with shadow and dimming effects.

4.10 Media & Attachments

Feature	Details
Voice recording	Real-time waveform, pause/resume, microphone permissions

Audio transcription	Speech Recognition framework for voice-to-text
Image embedding	JPG/PNG/HEIC picker, local disk storage, thumbnail cache
Web clips	URL metadata fetching (Open Graph), preview display
File attachments	Any file type, stored as references, reveal in Finder
File drop	Drag files onto editor to attach

4.11 Themes & Appearance

Feature	Details
Theme modes	System (follows OS), Light, Dark
Body font styles	Charter (serif), System (SF Pro), Monospaced
Line spacing	Compact (1.0x), Default (1.2x), Relaxed (1.5x)
Font size	Configurable base size; headings scale proportionally
Editor settings	Spell check, autocorrect, smart quotes, smart dashes toggles
Sort order	By date edited, date created, or title
Liquid Glass	Native glass materials with morphing, tinting, interactivity; .ultraThinMaterial fallback

4.12 Import & Export

Direction	Formats
Export	PDF (with embedded images), Markdown, HTML, Plain Text
Import	PDF, Markdown, HTML, Plain Text, RTF, DOCX, CSV

4.13 Security & Authentication

Feature	Details
Note locking	Per-note password protection
Auth methods	Custom password, system login password, Touch ID/biometric
Keychain storage	SHA256-hashed passwords with random salt
Auto-relock	5-minute unlock timeout with per-note session tracking

4.14 Apple Intelligence (macOS 26+)

Feature	Details

Summarization	2-4 sentence AI-generated summaries
Key points	3-7 extracted bullet points
Proofreading	Inline annotations with suggested replacements
Edit content	Custom instruction-based rewrites with before/after preview
Translate NEW	Selection-based translation with language picker and in-place replacement
Text generation NEW	Prompt-based text generation inserted at cursor position

4.15 UI Components & Interactions

Component	Details
Floating edit toolbar	Selection-aware formatting pills with submenus
Floating color picker	Palette + hex input + recent colors
Floating search	Glass-styled overlay search panel
Floating settings	Multi-tab preferences panel
Quick Look preview	PDFKit-based export preview with multi-page support
Sticker canvas	6 colors, resize/rotate/drag/z-index, click-to-place
Context menus	Pin, move, export, lock, archive, delete on right-click
Multi-selection	Cmd+click toggle, Shift+click range, anchor-based
Drag & drop	Notes to folders, notes to split panes, files to editor
Haptic feedback	Tactile responses on button taps and actions
Sidebar shortcut hints NEW	Keyboard shortcut hints appear on hover for sidebar menu items

5. Design System

Figma source: figma.com/design/BhVLOWG63LckTVCuO3q0Tv/Jot

5.1 Color Tokens

Token	Light	Dark	Usage
AccentColor	#2563EB	#608DFA	Primary interactive elements
MainColor	#FDBA74	#FDBA74	Brand accent
PrimaryTextColor	#1A1A1A	#FFFFFF	Body text
SecondaryTextColor	#737373	#A3A3A3	Supporting text

BackgroundColor	#FFFFFF (36%)	#0C0A09 (3%)	Window background
CardBackgroundColor	#FFFFFF (70%)	#1C1918 (70%)	Card surfaces
PinnedBgColor	#FEF08A	#854D0E	Pinned note accent
ButtonPrimaryBgColor	#1A1A1A	#FFFFFF	Primary buttons
SearchInputBgColor	#FFFFFF	#1C1918	Search field

5.2 Typography

Role	Font	Size	Weight
Body	Charter (or SF Pro / Mono per user pref)	14px	Regular
Heading 1	SF Pro Compact	32px	Bold
Heading 2	SF Pro Compact	24px	SemiBold
Heading 3	SF Pro Compact	20px	SemiBold
Metadata	SF Mono	11px	Regular
Micro	SF Pro	9px	Medium

5.3 Spacing & Radii

Scale	Values
Spacing	0, 2, 4, 8, 12, 16, 32, 48, 60 pt
Corner radius	0, 8, 12, 16, 20, 22, 24, 999 (capsule)
Sidebar row height	34pt
Concentric radii	Outer 16, padding 4, inner 12

5.4 Liquid Glass

Variants: `.regular` (toolbars, buttons), `.clear` (floating controls over media), `.identity` (conditional disable). Modifiers: `.tint(color)`, `.interactive()`. Shapes: Capsule, RoundedRectangle, Circle. Morphing via `.glassEffectID(id, in: namespace)` inside `GlassEffectContainer`. Fallback: `.ultraThinMaterial` on pre-26 systems.

6. Data Layer

6.1 SwiftData Schema

Entity	Core Fields

NoteEntity	id, title, content, createdAt, modifiedAt, isPinned, isArchived, isLocked, isDeleted, deletedDate, folderID, stickersData, webClipURL/Title/Description
FolderEntity	id, name, colorHex, isArchived, createdAt, modifiedAt

6.2 Rich Text Serialization

```
[[b]]bold[[/b]]  [[i]]italic[[/i]]  [[u]]underline[[/u]]  [[s]]strikethrough[[/s]]
[[h1]]Heading 1[[/h1]]  [[h2]]Heading 2[[/h2]]  [[h3]]Heading 3[[/h3]]
[[align:center]]centered[[/align]]  [[color|#FF5733]]colored[[/color]]
[[hl|#FFFF00]]highlighted[[/hl]]
[x] completed todo   [ ] incomplete todo
[[image|||filename.png]]  [[webclip|url|title|desc]]
[[callout|warning]]text[[/callout]]  [[code|swift]]let x = 42[[/code]]
[[table|...]]...[[/table]]  [[tabs|...]]...[[/tabs]]
[[cards|col1spec;col2spec]]col1content;;col2content[[/cards]]

Nesting order: align > heading/bold/italic > underline > strikethrough > color
```

6.3 Block Serialization Formats NEW

Block	Format
Cards	`[[cards|color:w:h+color:w:h;color:w:h]]content\t\tcontent;;content[[/cards]]` — `;` separates columns, `+` separates stacked cards, `;;` separates column contents
Tabs	`[[tabs|activeIdx|height|Name1\tName2|color1\tcolor2]]content1\t\tcontent2[[/tabs]]`
Card/Tab content	Uses `RichTextSerializer` tag format (same as main editor inline tags) for bold, italic, headings, block quotes, ordered lists, text color, highlight

6.4 Performance

Parameter	Value
Batch size	50 notes per fetch
Max load limit	500 notes
Auto-save debounce	3 seconds
Search debounce	250ms
Sidebar recompute	Only on note array changes

7. File Inventory

Largest Files

#	File	Lines	Responsibility
1	TodoEditorRepresentable.swift	9,004	NSTextView bridge to SwiftUI
2	ContentView.swift	5,460	Main layout, sidebar, split sessions
3	NoteDetailView.swift	1,925	Note editing screen
4	NoteTableOverlayView.swift	1,756	Table editor overlay
5	CardSectionOverlayView.swift NEW	1,656	Card section editor with rich text
6	TabsContainerOverlayView.swift UPDATED	1,377	Tabs container editor with rich text
7	FloatingSettings.swift	1,275	Settings panel
8	SimpleSwiftDataManager.swift	1,178	Data persistence manager
9	NoteImportService.swift	1,127	Multi-format import engine
10	TodoRichTextEditor.swift	904	Rich text editor component

Distribution by Layer

Layer	Files	Lines (approx.)
App (entry + main views)	3	7,400
Views / Components	37	24,000
Models	13	2,500
Utilities	19	11,800
Total	**109**	**45,703**

8. Development State

8.1 Timeline

Milestone	Date
First commit	August 5, 2025
Current date	March 26, 2026
Total commits	133
Primary contributor	Moheb Anwari

8.2 Recent Commits (March 2026)

Hash	Message
e12ef0a	feat: rich text editing in cards and tabs, AI routing, squiggly strikethrough, click-to-focus
f3089d1	feat: card section blocks with vertical stacking, magnetic snap, and code audit cleanup
68718e7	feat: sidebar shortcut hints on hover, split picker dim removal, toolbar gap reduction
2395c60	fix: Body style reset, undo animation, update panel image alignment
cf80276	fix: macOS 14 compatibility, block quote crashes, theme transition, sidebar UI
ddcd838	fix: squiggly strikethrough vertical alignment on wrapped todo lines
8a7481f	feat: AI tools (translate, text gen, edit content), block protection, floating toolbar fix
96e6221	feat: callout block UI redesign + floating sidebar preview card fix

8.3 Current Phase

Pre-release stabilization with active feature development. All core features implemented. Recent work focuses on card section blocks, rich text in overlay editors (cards + tabs), AI tool integration, squiggly strikethrough polish, and a 14-batch code audit cleanup.

9. Feature Roadmap

13 planned features across 6 batches. Issues tracked in Linear (DES-265 through DES-277).

Batch 1 — Non-Negotiable, High Priority

Feature	Linear	Description
Comprehensive Keyboard Shortcuts	DES-265	Cmd+N, Cmd+Shift+N, arrow-key sidebar nav, Cmd+Backspace delete, Cmd+F find
Toast-Based Undo	DES-266	5-second toast after delete/move/archive/pin with undo action

Batch 2 — Non-Negotiable, High Priority

Feature	Linear	Description
Spotlight Integration	DES-267	macOS Spotlight search indexing + deep link navigation
Printing	DES-268	Cmd+P with NSPrintOperation, styled PDF output

Batch 3 — Nice-to-Have, High Priority

Feature	Linear	Description

Share Extension	DES-269	macOS share sheet target for capturing content from other apps
Siri Shortcuts & App Intents	DES-270	Voice commands and automation shortcuts for note creation/search

Batch 4 — Non-Negotiable, Medium Priority

Feature	Linear	Description
Auto Local Backups	DES-271	Periodic automatic backups with restore UI
Note History / Versioning	DES-272	Timeline viewer for note revision history

Batch 5 — Nice-to-Have, Medium Priority

Feature	Linear	Description
Widgets	DES-273	WidgetKit widgets for pinned/recent notes on desktop
Reminders / Due Dates	DES-274	Lightweight date reminders on notes

Batch 6 — Nice-to-Have, Medium Priority

Feature	Linear	Description
Global Quick Note Hotkey	DES-275	System-wide keyboard shortcut for instant note capture
Smart Folders	DES-276	Rule-based auto-populating folders
Templates	DES-277	Pre-structured note templates + save-as-template

Decided Against

Feature	Reason
Backlinks Panel	Not aligned with app simplicity goals
Focus Mode	Low user demand
Quick Capture from Clipboard	Covered by Global Quick Note

10. CI/CD

Workflow	Trigger	Purpose
claude-code-review.yml	Pull request events	Automated code review via Claude Code Action
claude.yml	@claude mentions in issues/PRs	AI assistant for issue triage and PR tasks

Build Commands

```
xcodebuild -project Jot.xcodeproj -scheme Jot -configuration Debug build
xcodebuild -project Jot.xcodeproj -scheme Jot -destination 'platform=macOS' test
xcodebuild -project Jot.xcodeproj -scheme Jot clean
```